06 · L2 — Decision Logic

> **Assignment:** *“Decision logic: specify decision logic. This can be done diagrammatically (BPMN), or in a decision table (e.g. as per WHO SMART Guideline DAK)."*

We document the decision logic **in both forms**: as a decision table here and diagrammatically in the detailed app process. The table states *what* is decided, the diagram shows *in which order*.

6.1 Object of the Decision

Decision ID	`DEC-01 Wundstatus`
Trigger	Completion of a daily wound check
Business Rule	Classify the current wound findings into one of four urgency levels and provide a concrete recommendation for action
Outputs	🔴 RED (emergency) · 🟠 ORANGE (suspected infection) · 🟡 YELLOW (abnormal) · 🟢 GREEN (normal)
Additional output	`DEC-02 Hinweise` — non-exclusive informational messages that fire in addition to the classification

---

6.2 Evaluation Principle: worst-first

The rules are evaluated **in a fixed order from severe to mild**. The **first matching rule wins**, and the cascade ends. Informational rules (I1–I5) are exempt from this and fire in addition.

```
 R1 → R2 → R3           🔴 RED
R9 → R4 → R5 → R11 → R12   🟠 ORANGE
R6b → R6a → R7 → R10       🟡 YELLOW
R8 (Catch-all)             🟢 GREEN
────────────────────────────────
I1 … I5                    ℹ️ additional, non-exclusive
```

Rationale: For a triage tool in the hands of patients, false reassurance weighs more heavily than a false alarm. The cascade guarantees that reassuring findings can never override a warning finding. In exchange, we deliberately accept a higher false-alarm rate — see 12 Impact.

6.3 Decision Table — Wound Status

Rules R6a/R6b, R9–R12 and I5 were added in L3 v0.3.x as a result of the defects found during testing — see 10 QA.

🔴 RED — Emergency

ID	Condition	Output	Rationale	L1 reference
R1	`temp >= 39` **OR** `chills = 1`	**EMERGENCY** — High fever or chills: contact your practice or the emergency service **immediately**.	Systemic signs of infection have the highest priority. Chills are a sepsis warning sign and may precede the fever	CDC (fever >38 °C as a criterion) · sepsis red flags

ID	Condition	Output	Rationale	L1 reference
R2	`red_streak = 1` **OR** `redness_extent = 4`	**EMERGENCY** — Erythema is spreading / red streak: seek medical assessment today (suspected lymphangitis).	A red streak tracking away from the wound is a classic sign of lymphangitis and requires prompt antibiotic therapy	NICE NG125 §1.4.9 (cellulitis)
R3	`wound_open = 2` **OR** `bleeding = 2` **OR** `wound_state = 4`	**EMERGENCY** — Open wound, pus or heavy bleeding: contact your practice / emergency service immediately. Cover the wound with a clean compress.	Wound dehiscence and purulent discharge are criteria of deep incisional SSI. The wound appearance "open/pus" **always** escalates, even if all other entries are unremarkable	CDC deep incisional SSI · Southampton IV–V

🟠 ORANGE — Suspected Infection

ID	Condition	Output	Rationale	L1 reference
R9	`wellbeing = 2` **OR** `nausea_vomiting = 2`	**SUSPECTED INFECTION** — You feel distinctly unwell: contact your practice today, even if the wound looks unremarkable.	A severe subjective feeling of illness may **precede** measurable fever. The element was collected but originally not used in any rule — see Bug B-02	Systemic signs of infection
R4	`secretion = 3` **OR** `secretion_smell = 1`	**SUSPECTED INFECTION** — Purulent or foul-smelling discharge: contact your practice today for an appointment.	Purulent discharge is a **main criterion** of superficial incisional SSI ("purulent drainage from the superficial incision")	**CDC** superficial incisional SSI · Southampton IV
R5	`fever = true` **OR** (`pain >= 7` **AND** `pain_trend = 2` **AND** `days_post_op >= 3`)	**SUSPECTED INFECTION** — Fever or increasing severe pain: contact your practice today.	Pain should be decreasing from day 3 onwards. An increase is a warning sign — the CDC explicitly names *"new or worsening localized pain"*	**CDC** · NICE NG125
R11	`secretion >= 1` **AND** `discharge_duration_days >= 3`	**SUSPECTED INFECTION** — Persistent wound discharge for several days: contact your practice today.	Southampton lists "prolonged discharge > 3 days" as a separate sub-grade (IIId) — persistent discharge requires treatment even in the absence of pus	**Southampton IIId**
R12	`classification = 'yellow'` **AND** `last_classification = 'yellow'`	**SUSPECTED INFECTION** — Second abnormal day in a row: contact your practice today.	A single abnormal day is tolerable, a persisting abnormal finding is not. Originally specified as "deterioration vs. yesterday" and therefore unreachable — see B-08. Requires two-pass evaluation (B-07)	CDC/NHSN — new or worsening · persistence criterion

🟡 YELLOW — Abnormal

ID	Condition	Output	Rationale	L1 reference
R6b	`southampton_grade = II` **OR** `swelling = 3`	**ABNORMAL** — Distinct signs of inflammation: observe the wound closely and contact your practice if it has not improved by tomorrow.	**Applies from day 1.** Pronounced signs of inflammation are not normal even early after surgery — correction of Bug B-01	Southampton II · CDC (erythema, swelling, heat)

ID	Condition	Output	Rationale	L1 reference
R6a	(`southampton_grade = I`) AND `days_post_op >= 3`	**ABNORMAL** — Increasing signs of inflammation: observe and check again tomorrow.	Mild erythema/swelling is a physiological inflammatory response during the first 2–3 days. From day 4 onwards it is no longer	Wound healing phases · Southampton I
R7	`wound_open = 1` OR `numbness = 1` OR `mobility_change = 1`	**ABNORMAL** — Small wound gap, numbness or restricted movement: raise this at your next follow-up appointment; contact your practice if it worsens.	Minor dehiscence and nerve irritation are not emergencies, but they must be documented and observed	Follow-up care standards
R10	`bleeding >= 1` AND `anticoag = 1`	**ABNORMAL** — Bleeding while on blood thinners: observe the dressing and report any increase.	Bleeding must be assessed differently under anticoagulation than without it. The element `anticoag` was collected but originally not used — see Bug B-05	Bleeding assessment

🟢 GREEN — Normal

ID	Condition	Output	Rationale
R8	*(Catch-all — none of the rules above applies)*	**ALL CLEAR** — Wound healing is progressing normally. Perform the next check tomorrow.	Explicit else rule. Guarantees that **every** combination of inputs receives a defined result — there is no hole in the logic

---

6.4 Decision Table — Notices *(non-exclusive)*

These rules fire **in addition** to the classification and may overlap with one another.

ID	Condition	Output
I1	`risk_count >= 2`	You have several risk factors for impaired wound healing — please attend all of your follow-up appointments.
I2	`meds_taken = 0`	Please take your prescribed medication — it protects wound healing.
I3	`smoker = 1`	Smoking slows wound healing — now would be a good moment for a break.
I4	`crust = 1`	A scab (crust) is a normal sign of healing — do not scratch it off, keep the wound dry.
I5	`bmi >= 30` OR `wound_class >= 3`	Your procedure carried an elevated risk of wound infection — pay particular attention to the daily check.

I4 is a deliberate design decision. A scab is a normal sign of healing, but patients frequently interpret it as a problem (see Scenario A). Instead of omitting the question, we collect it and actively give the all-clear. At the same time, worst-first applies: if purulent discharge is present simultaneously, R4 wins — the scab notice then appears *alongside* the warning, not instead of it. Tested in T10.

Evaluation in two passes

Rule R12 references `classification` — its own result. The rules are therefore evaluated **twice**: the first pass produces a provisional classification, which is fed back into the scope; the second pass may then fire R12. **The worse of the two results wins**, so a second pass can only escalate, never de-escalate — worst-first applies across passes as well.

This mechanism was added after bug B-07, in which R12 could never fire because of a circular dependency.

6.5 Completeness and Consistency Check

Check	Result
Every combination of inputs reaches a defined result	✅ guaranteed by catch-all R8
Prioritisation defined for multiple matches	✅ worst-first cascade, order documented
Every data element with warning relevance feeds into at least one rule	✅ after fixing B-02 and B-05
Every rule references an L1 source	✅ except R8 (catch-all, by definition without a source)
Every rule has at least one test case	✅ see 10 QA Testing
No unreachable rule	✅ after fixing B-03 (`red_streak` no longer gated) and B-08 (R12 rewritten as a persistence rule)

Truth Table for the Critical Rule Group

Systemic signs (R1, R5, R9) — the group with the highest potential for harm in the event of misclassification:

`temp`	`chills`	`wellbeing`	Result	Rule
39.2	0	0	🔴 RED	R1
38.0	1	0	🔴 RED	R1 *(chills override temperature)*
39.0	0	0	🔴 RED	R1 *(threshold inclusive)*
38.5	0	0	🟠 ORANGE	R5
38.0	0	0	🟠 ORANGE	R5 *(threshold inclusive)*
37.4	0	2	🟠 ORANGE	**R9** *(without R9 this would be GREEN → Bug B-02)*
37.4	0	1	🟢 GREEN	R8
36.9	0	0	🟢 GREEN	R8

6.6 Thresholds

All threshold values are stored as named constants in the L3 (`meta.thresholds`) and are referenced by the rules — **not** hard-coded into the rule expressions. A change in a guideline is therefore a change to a single line.

Constant	Value	Meaning	Origin
`fever_orange`	38.0 °C	Fever threshold for suspected infection	CDC deep incisional SSI
`fever_red`	39.0 °C	Fever threshold for emergency	Group decision → C-07
`pain_high`	7	Severe pain (NRS)	NRS convention
`normal_healing_days`	3	End of the physiological inflammatory phase	Group decision → C-03
`discharge_prolonged_days`	3	Persistent discharge	Southampton IIId
`bmi_risk`	30	Obesity as an SSI risk factor	WHO definition

> **Demo moment for the presentation:** change `fever_orange` from 38.0 to 37.5, reload the app — the same test case now comes out ORANGE instead of GREEN. Without a single line of code. That is the proof that the separation of layers is real.

Related documents: 03 L1 · 05 Data Dictionary · 07 Challenges · 10 QA